

Interviews and Interviewers II:

Interviewing Techniques and Interviewer Effects

Fundamentals of Data Collection II

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Interviewers and Interviews: Outline

■ Obtaining interviews

- Recruiting sample members
- Within household selection
- Proxy respondents

■ Interviewing Technique

- Standardization Debate: Wording vs. Meaning
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- Formal vs. Personal Style
- Rapport between Interviewers and Respondents
- Increasing Respondent Motivation

■ Interviewer effects

- Variance: Interviewer Behavior
- Bias: Fixed Attributes

Part I

Part II

Personal vs. Formal Interviewing

- Two approaches to interviewing explored by Dijkstra (1987): “Formal” and “Personal” interviewing styles
 - Formal: business-like, no small talk
 - Personal: connect with *R* on an individual basis (e.g., “Good for you” or “Me too”)
 - Note: ≠ conversational vs. standardized interviewing distinction
- Both approaches require learning basics of interviewing, e.g., reading questions as worded, probing, as well as specifics of style
 - Formal style includes “unlearning” person-oriented behavior

Personal vs. Formal Interviewing (2)

- Dijkstra randomized *Iwers* to formal or personal interviews and asked:
 - will Personal style motivate *Rs* to answer as accurately as possible or will it lead them to try to please *Iwer*, i.e., to ingratiate themselves
- and measured impact on accuracy and social desirability
- Accuracy:
 - asked *Rs* draw map of their hometown
 - coded for proportional correctness and inclusion of key landmarks, etc.
- Social Desirability:
 - Administered shortened version of Marlowe-Crowne impression management scale
 - Higher scores indicate tendency toward more socially desirable responses

Accuracy of Drawings

Higher scores indicate greater accuracy (max=37)

Respondent Gender		
Interviewing Style	Male	Female
Personal	16.3	14.5
Formal	14.6	12.9

Social Desirability

Higher scores indicate more social desirability

Respondent Gender		
Interviewing Style	Male	Female
Personal	11.4	12.5
Formal	12.1	13.0

- Rs in personal style gave reliably more irrelevant statements (ingratiation)

Responses to Two Questions by Style of Interviewing

		Formal	Personal
After moving to this neighborhood, did you try to make contact with people living here?	Socially desirable (% yes)	36	33
	Socially undesirable (% no)	46	64
	DK	18	3
		Formal	Personal
Are there people living in this neighborhood who you do not like?	Socially desirable (% no)	57	61
	Socially undesirable (% yes)	33	34
	DK	10	5

van der Zouwen *et al.*, (1991) “Studying Respondent-Interviewer Interaction: The Relationship Between Interviewing Style, Interviewer Behavior, and Response Behavior”

Rapport and Disclosure

- Possible increased rapport will inhibit disclosure of sensitive info because *R* wishes to maintain *Iwer*'s favorable impression of them
- Sun et al. (2021) manipulated question sensitivity (low, moderate, high) and measured rapport between *Iwer* and *R*
 - Sensitivity of Qs independently rated
 - Collected *Rs*' ratings of rapport with *Iwer* immediately after interview
 - Found more disclosure when *Rs* rated rapport high suggesting rapport promotes candor more than it promotes impression management
 - Mechanism seems to be reduced social distance between *Iwer* and *R* created by increased rapport

Rapport and Acceptable Responses: *Iwer's* Nonverbal behavior

- Welles, Sun & Miller (2022) examined relationship between “adequate responses”* and non-verbal behavior of 4 NSFG** *Iwers* known to affect/reflect rapport: smiling, nodding, direct gaze
- Adequate responses more likely when *Iwer* engaged in the three non-verbal behaviors
 - Direction of causality unclear but implication is rapport => better responses
- Relationship unaffected by degree of standardization in interaction
 - Suggests may be possible for *Iwers* to build relationship with *R* using nonverbal behaviors without deviating from standardized techniques

* responses that matched one of the expected response categories and required no additional probing

**National Survey of Family Growth

Summary: Personal vs. Formal Interviewing; Effects of Rapport on Disclosure and Acceptable Responses

- Personal style motivates respondents to give accurate and candid responses relative to formal style
 - Minor increase in ingratiation
- Seems to work by increasing rapport between *I* and *R*
 - Note: Personal $\neq$ Conversational Interviewing
- Increased rapport seems to increase disclosure of moderately and extremely sensitive behavior, presumably by reducing social distance
- *I*ver non-verbal behaviors associated with increased rapport (smiling, nodding, direct gaze) are more common when responses acceptable

Motivating and “Training” Respondents

- Cannell, Miller & Oxenberg (1987) developed an interlocking set of techniques to positively affect how *Rs* approach their task
- Experimental treatment bundles together three techniques intended to improve quality of responses, compared to none of these in control group
 1. *Commitment*
 - Verbal agreement by respondent to give accurate and complete information
 2. *Instructions*
 - Statements read by *I* for specific Qs: “For the next few questions, you may need to take time to think carefully before you answer to be the most accurate.”
 3. *Feedback*
 - Both positive (“I see, this is the kind of exact answer we need.”) and negative (“You answered that quickly. Are there any days you might have overlooked?”)

Percentage Reporting at Least One Health Event by Experimental Condition

	Control	Experimental
Illness-related...		
Days in Bed	7.8%	10.9%*
Loss of Work Day	6.7	9.4*
Days of Reduced Activity	8.8	12.0*
Dental Visits	6.7	7.4
Doctor Visits	17.2	17.2
Acute Conditions	15.0	18.1*

* $p < .05$ Cannell et al., (1987)

- Assuming more reports reflects more complete recall, experimental approach (commitment, instructions, and feedback) improves data quality for several questions

Commitment in Web Surveys

- *Rs* can be asked by an *Iwer* to commit to conscientious responding but can also be asked in web questionnaire
- Conrad et al. (2017): online *Rs* asked to commit; less likely to speed and (in one experiment) to straightline
- Cibelli-Hibben et al., (2020): online *Rs* asked to commit; less likely to skip items, more likely to check records, responded more slowly, reported income more accurately than control

Interviewing Technique Summary

- Interviewing technique can have a major effect on response quality
- Interviewers must use judgment and discretion, i.e., cannot behave robotically, when using any technique
 - Irrespective of techniques, less improvisation conducting interview, than in recruiting sample members

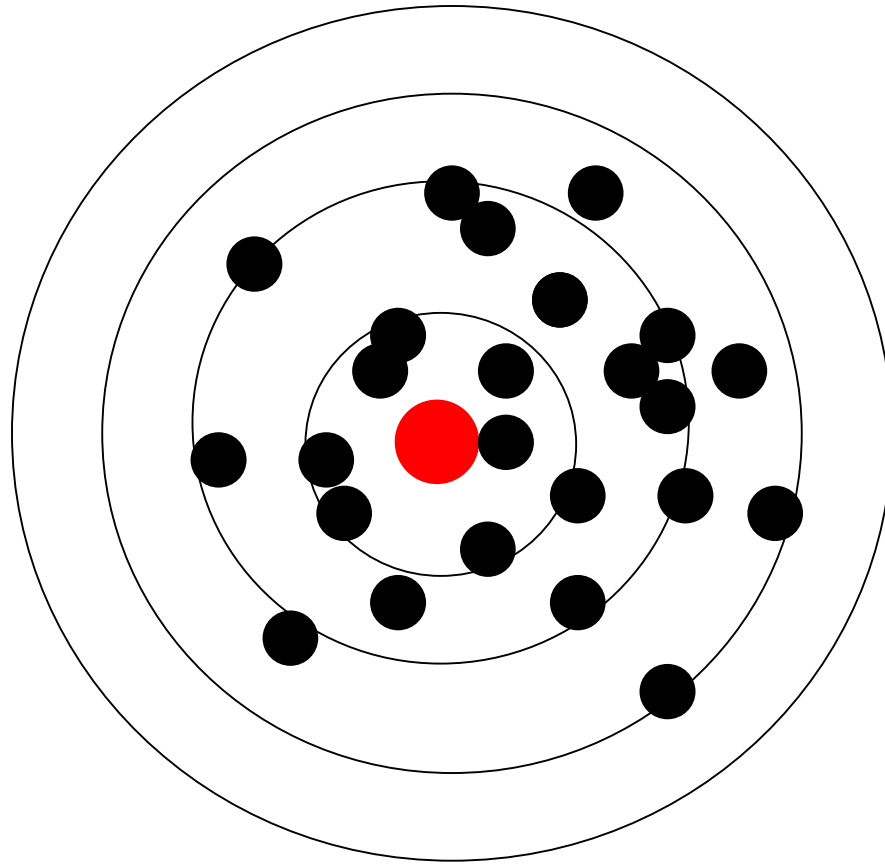
Interviewer Effects

- Interviewers add value to the survey enterprise
 - e.g., increase response rates, explain Q meaning to R , conduct within-HH sampling
- But can also **change** [introduce] error – *interviewer effects*
- 2 main kinds of interviewer effects (coverage error, nonresponse error, sampling error and **measurement error**), due to:
 - *I*nterviewer behavior (administering questionnaire); affects *I*-related **variance**; measured for individual questions, i.e., not entire q'aire
 - *I*nterviewer fixed attributes, e.g., race, gender, age; associated with directional error, i.e., **bias**, for questions related to attribute, e.g., gender of interviewer effects for Qs on gender-relevant topics

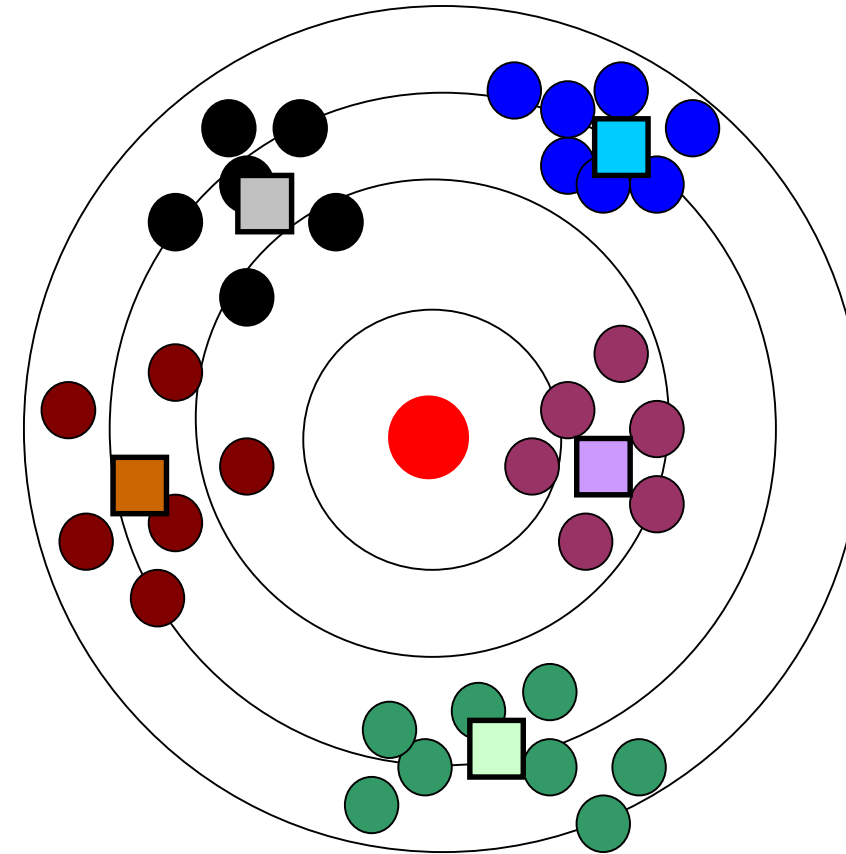
Interviewer Effects: Behavior

- Foundation of Standardized Interviewing (SI) is that if *Interviewers* are interchangeable, response variance due to *Interviewers* should be minimized
- But lots of evidence that SI not perfectly implemented by all *Interviewers*
- Differences in behavior between one *Interviewer* and the next likely to inflate *Interviewer*-related variance (ρ_{int})
 - Different distribution of answers collected by different *Interviewers*
 - Reduces precision (increases confidence interval) of survey estimates
 - Measuring ρ_{int} requires random assignment of *Respondents* to *Interviewers*; otherwise, diff's. could reflect truth
 - If ρ_{int} high, implies some responses less accurate than others — although do not know which responses are less accurate
- SI conducted to reduce diff.'s in *Interviewer* behavior and so reduce ρ_{int}

Interviewer Variance



Without interviewer variance



With interviewer variance

- Bull's eye (red filled circle) = true value (sample mean) for one Q
- Black shots = responses without clustering by *lwer*
- Color-coded shots = responses clustered by *lwer* (same color square)

Estimating *I*wer Variance

- Kish (1962) introduced ρ_{int} (or $\rho_{\text{ho}_{\text{int}}}$) to capture the correlation between interviewers and the answers they elicit to a *particular Q* or *I*wer clustering
 - sometimes called *intraclass correlation*

$$\rho_{\text{int}} = \frac{\text{between} - \text{interviewer variance}}{\text{between} - \text{interviewer variance} + \text{within} - \text{interviewer variance}}$$

- Between-interviewer variance is ideally zero
 - But never is!
 - Assumes random assignment of *I*'wers to *Rs*, , i.e., *interpenetrated* design
- Within-interviewer variance is (essentially) range of values for the variable (question) of interest; measures item's ability to discriminate among *Rs*

Kish's formulation of ρ_{int}

$$\rho_{int} = \frac{\left(\frac{V_a - V_b}{m} \right)}{\left(\frac{V_a - V_b}{m} \right) + V_b}$$

- V_a = between subjects mean square error in a one-way ANOVA with interviewers as the subjects
- V_b = within subjects mean square error in ANOVA
- m = total number of interviews conducted by the interviewer

Interviewer Behavior and Interviewer Effects

- Particular *lwer*-behaviors may affect ρ_{int}
- Mangione, Fowler & Louis (1992) coded *lwer*-behaviors in audio-recorded interviews
- Computed correlation between each behavior and ρ_{int}
- Most correlations very small but those involving probes are larger and significant
- Probes are unscripted despite being central to standardized interviews

Significant Correlations between ρ_{int} and specific interviewer behaviors

Correct Probe	.23
Directive Probe	.20
Failed to Probe	.49

Source: Mangione, Fowler & Louis (1992)

Interviewer Effects for Different Types of Questions

- O' Muirheartaigh & Campanelli (1999) observed significant ρ_{int} in:
 - Attitude questions (26%) with Likert scales (33%)
 - Factual questions (26%)
- *Interviewer* effects inflate variance about as much as geographical clustering
- Schnell & Kreuter (2005) examined 108 questions about crime; *Interviewer* effects greater for
 - Sensitive than nonsensitive Qs
 - Nonfactual than factual Qs
 - Open than closed Qs
 - Difficult than easy Qs

Impact of *lwer* Effects on Precision of Estimates

- How much does ρ_{int} , i.e., *lwer* clustering, inflate overall variance?
- Can calculate inflation by treating all other variance as a constant and adding ρ_{int} multiplied by the mean number of interviews for each *lwer*
$$\text{deff}_{\text{int}} = 1 + \rho_{\text{int}} (m-1)$$
- Design Effect due to *lwers* :
 - inflation of overall variance due to ρ_{int}
 - m = average interviewer workload (# interviews)
 - m moderates (i.e., amplifies or diminishes) impact of ρ_{int}
- Sometimes researchers use the square root of the Design Effect, called the Design Factor: (deft) is $\sqrt{\text{deff}}$
 - more useful for adjusting standard errors

Effect of *lwer* Workload on Precision of Estimates

$$Deft_{int} = \sqrt{1 + \rho_{int}(m - 1)}$$

With a ρ of .02 and mean # interviews (m) of 10,

$$\begin{aligned} Deft_{int} &= \sqrt{1 + 0.02(10 - 1)} \\ &= \sqrt{1.18} \\ &= 1.087 \end{aligned}$$

If the average workload (m) were increased to 50,

$$\begin{aligned} Deft &= \sqrt{1.98} \\ &= 1.41 \end{aligned}$$

Relationship between ρ_{int} , workload, and design effects

	Telephone Methodology	Health in America	Consumer Attitudes
Response Rate	59%	80%	74%
Number of interviewers	37	33	26
Average workload	41.3	58.1	14.1
Mean values:			
ρ_{int}	.0089	.0018	.0067
deff_{int}	1.36	1.10	1.09

Source: Table 1 in Groves and Magilavy (1986)

- Assuming a fixed budget for interviewing, more *interviewers* each conducting fewer interviews, dilutes the effects of idiosyncratic *interviewer*-behavior

Impact of *lwer* Effects on Precision

- ρ_{int} permits computation of *design effects* due to *lwers*
- Design Effect (*deff*) makes it possible to see effect of *lwer* variance on overall variance
- Effective Sample Size quantifies impact of *deff* on power and precision:

$$\frac{n}{deff}$$

- e.g., if $n = 1000$ and $deff = 1.26$, effective sample size = 794

Response Validity versus *Iwer* Variance

- *Iwer* variance (i.e., ρ_{int}) is correlation between *Iwers* and responses they elicit
 - Although Standardized *Iwers* should be interchangeable, may be association between *Iwer* and responses they elicit
 - If so, *suggests Iwer* is affecting answers; *implies* variation in data quality
- But we can't know for sure because ρ_{int} is an indirect measure of data quality
 - Possible for ρ_{int} to be low (good) and response validity to be low (bad) if all interviewers consistently collect the wrong answer
 - e.g., *Rs* asked if have recently visited doctor; survey intended to measure MD visits, but *Iwers* not instructed to or not permitted to define “doctor”; many *Rs may* include visits to non-MDs, e.g., PAs, podiatrists, chiropractors
 - ρ_{int} would be low (uniform response distributions across interviewers) but accuracy would also be low (more “yes” answers than actual MD visits)
- Mean Square Error (MSE) is composite of *Iwer* variance (ρ_{int}) and (in)accuracy (squared bias)

Does Conversational Interviewing increase *lower* variance?

- Proponents of SI argue that departure from scripted interaction will increase interviewer variance (ρ_{int})
- If so, does the increase in interviewer variance undermine the advantage of CI for data quality
- West, et al. (2018) tested by calculating ρ_{int} (IIC) and data quality
 - Data quality measured by (1) Comparing answers to administrative records (accuracy/bias) and (2) Response distributions in conjunction with definitions (inclusive* vs. exclusive** defs)
 - calculated ρ_{int} (IIC) for all Qs for CI and SI
 - Where possible calculated MSE

*e.g.: household member *includes* child away at college

**e.g: moving expenses *do not include* DIY expenses such as fuel, food, and lodging

Does CI increase *lower* variance? (2)

- For 36 of 55 Qs (65.5%), interviewing technique did not significantly affect IICs or response distributions
- IICs significantly higher for one Q for CI than SI, marginally higher in 4
- Administrative data, required for MSE, available for 1 of the 5 Qs:
 - *longest uninterrupted period of employment in previous 20 years*
- Based on admin data, CI led to significantly higher accuracy than SI
 - leading to smaller MSE^* (better quality) for CI than SI despite increased IIC
 - CI produces wider confidence interval around more accurate point estimate than SI
- For 10 Qs in which estimates (e.g., means, %s) between techniques differed significantly, the results were more accurate (consistent with definitions) for CI than SI

*MSE is squared bias (inaccuracy) plus variance (including IIC)

Does CI increase *lower* variance? (3)

- Not much, if at all
 - For most Qs, IIC does not differ significantly between SI and CI
 - On the rare occasions when there are significant differences in IIC, CI improved accuracy more, offsetting higher IICs, resulting in smaller MSEs
- CI improves quality of reporting relative to SI consistent with previous findings *without notably increasing interviewer effects*
- *Note:* Duration of CIs ~1 min more than SI (ns)

Summary: *Iwer* Effects Attributed to *Iwer*-Behavior

- Probing is key behavior related to ρ_{int} – the only unscripted behavior in SI
 - But no greater in CI (where more unscripted *Iwer*-behavior) than SI, despite increased response accuracy in CI
- Impact of ρ_{int} on precision is “diluted” by reduced average workload (fewer interviews per *Iwer*)
- ρ_{int} generally not a direct measure of response accuracy
 - measures variance (correlation between *Iwers* and answers); if *Rs* randomly assigned to *Iwers*, *Iwer-related* error likely reflects measurement error
- Response accuracy, when available, is a more definitive measure

lwer Effects: Enduring Characteristics

- Certain interviewer characteristics, primarily race and gender (but also age, class, ethnicity) have been shown to affect responses
 - systematically shifting answers in one direction – bias
- But only for Qs for which the attribute is relevant -- not for other Qs
 - e.g., race of interviewer only affects answers to Qs that concern race such as racial attitudes
- Distinct from literature on *lwer* variance

Gender of Interviewer

- Kane & Macaulay (1993) report that both male and female *Rs* gave more egalitarian answers to female than male *lwers*... but for different items
 - More men reported sharing childcare when *lwer* was female
 - More women supported feminist policy when *lwer* was female
- Huddy et al. (1997) report numerous gender of interviewer effects in two pre-election surveys
 - Male and female *Rs* show gender of *lwer* effects for same questions

Gender of Interviewer

- Huddy et al. (1997)

	F-M diff in rating (10-point scale)	
Year	1991	1993
Feminist Identity	.76***	
Activity on Women's Issues	.50*	
Pro-Collective Action	.48*	.69**
Egalitarian Roles	.35*	
Pro-Choice		.82***
Anti-Sexual Harrassment		.63**
Pro-Affirmative Action	.84**	

* $p < .1$ ** $p < .05$ *** $p < .01$

Race of Interviewer (ROI) Effects

- Analogous to gender of interviewer effects
 - Only observed for Qs for which race is relevant
 - Even for those items, effects only observed some of the time
- Effects generally larger in face-to-face studies but ROI effects reported in phone interviews

Race of Interviewer Effects: White Respondents

Category	Black interviewers	White interviewers
Would not mind if relative married a Negro	71.7%	25.5%
Believes Negro and white students should go to same school	90.7%	56.0%
Would not be disturbed if Negro of same class moved into block	100.0%	68.6%
Believe Negro and white children should play together freely	92.5%	84.0%

Source: Table 1 in Hatchett and Schuman (1975).

Race of Interviewer Effects: Black Respondents

		% in Category	
Question	Category	White lw'r	Black lw'r
Can trust white people?	Trust most whites	35	7
Negro parents work best with Negro teacher?	Yes	14	29
Favorite entertainers?	Named only black entertainers	16	43

Schuman & Converse, 1971

ROI Effects in Telephone Interviews

- Davis (1997) reports higher correlation between support for Jesse Jackson and racial consciousness for Black than White *Interviewers*
- Davis and Silver (2003)
 - Black *Rs* answered fewer (3.4 vs 2.8) knowledge questions correctly – stereotype threat – when perceived *Interviewer* to be Black
 - But no significant difference based on *Interviewers*' self-reported race
 - For White *Rs* no effect of perceived or self-reported *Interviewer* race
 - *Rs*' ability to judge race varies widely: estimates of “correct” race identification 14%-82%

*I*wer Effects Due to Fixed Attributes: Summary

- Iwer's attributes such as race and gender can lead to biased responses when topic of Q relevant to attribute
 - Stronger in person but effects of (perceived) race observed on phone

Interviewers: Summary/Conclusions

- Largest cost component of surveys involving interviewers
- Potentially large impact on survey errors across tasks
 - Recruiting, Respondent Selection
 - Data Collection
- *Interviewer* effects: behavior, especially probing; fixed attributes e.g., race, gender
- All Interviewing Techniques have Pros and Cons:
 - SI weak at resolving misconceptions; CI can improve response accuracy but leads to longer interviews; little difference in interviewer variance
 - Personal style can motivate *Rs* but can lead to ingratiation; increased rapport seems generally to improve data quality
 - Commitment seems to improve data quality (with new evidence in web surveys)

Interviewers: Summary/Conclusions (2)

- Despite growth of self-administered data collection, *Interviewers* will remain vital component of survey data collection
- But probably in more targeted ways
 - e.g., difficult response tasks, exit polls, non-response follow-up to mailed survey invitations, need for physical/bio measures, samples of hard-to-reach populations, etc.